

Lab 05 - Combinatorial Logic

In this lab, you've learned real world applications of digital logic, as well as how to assemble your own Verilog modules. In addition, you've learned how the constraints file maps your inputs and outputs to real pins on the FPGA.

Rubric

Item	Description	Value
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Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Name

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Lab Summary

In this lab we learned how to build our own circuit and top files, write in verilog, and analyze and understand the constraint file. We continued to practice finding minterms and maxterms, creating k-maps, and finding the most reduced form of circuits.

Lab Questions

1 - Explain the role of the Top Level file.

The role of the top level file is to manage circuits and ports from a high level. The top file is the root of the hierarchy that defines the structure of the circuit. Since the top file is the highest priority, it can manage and call all the other files in the project including the constraints file.

2 - Explain the function of the Constraints file.

The constraints file serves to provide instructions about hardware. It manages the ports, pins, timing, and power to ensure that the physical board can run the files. The top file gets these specifications from the constraints file.

3 - Was the selection of Minterm and Maxterm correct for each circuit? What would you have chosen?

Either was correct because they yield the same outcome, however if we were able to choose one, choosing minterms for Circuit A would've been easier. Circuit A only had 4 outcomes that

were 1, focusing on those values would have been quicker than focusing on the 12 zeros. Circuit B did not matter because there were an equal number of 1's and 0's in the output.